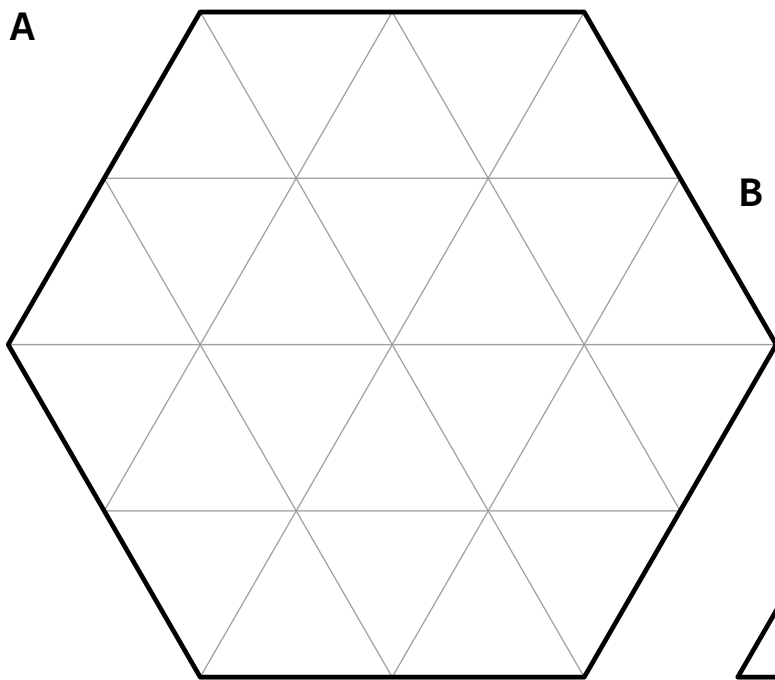


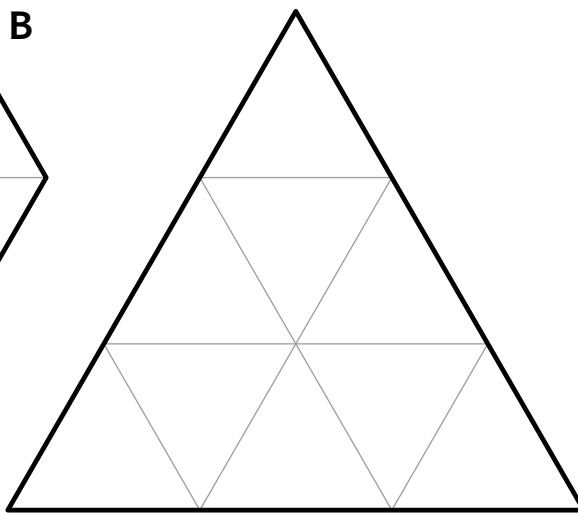
Use the small pattern blocks. Blocks lie flat inside the outline and fit the lines of the grid. They may not overlap or stick out past the edge, and they may not cover a black triangle. To cover a board means to put a block over every small triangle of it.

Problem 1: Cover each board with blue rhombuses. Draw your covering on the board. If a board cannot be covered, write “cannot” under it.

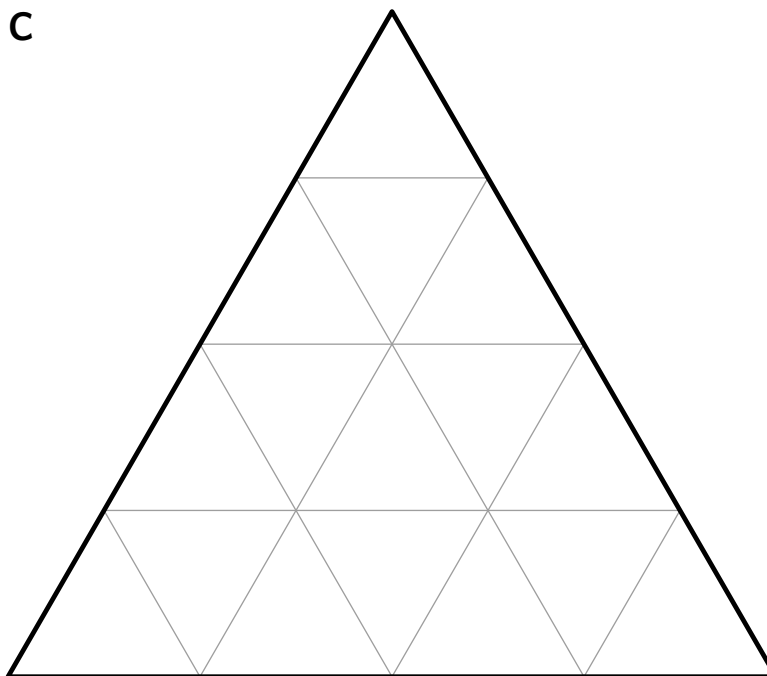
A



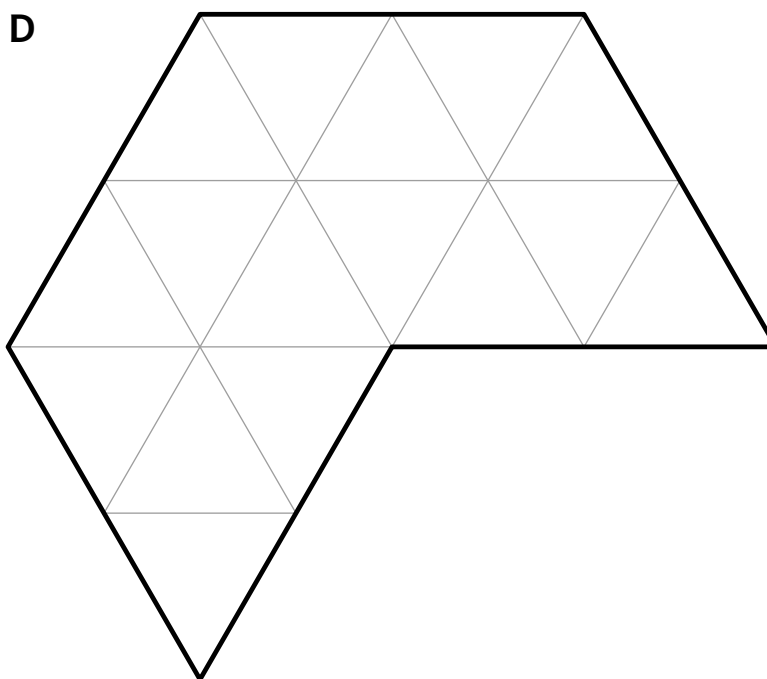
B



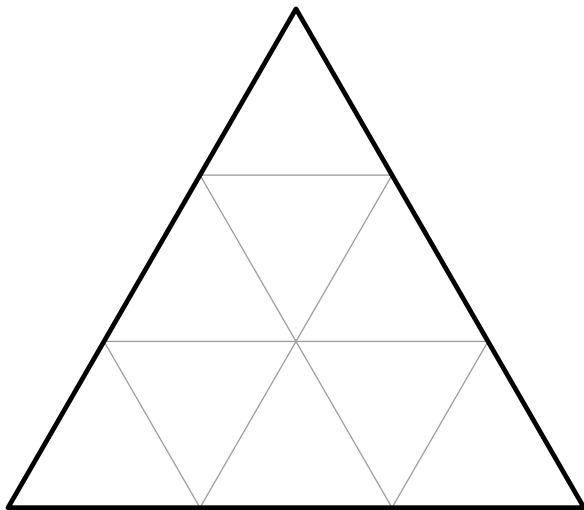
C



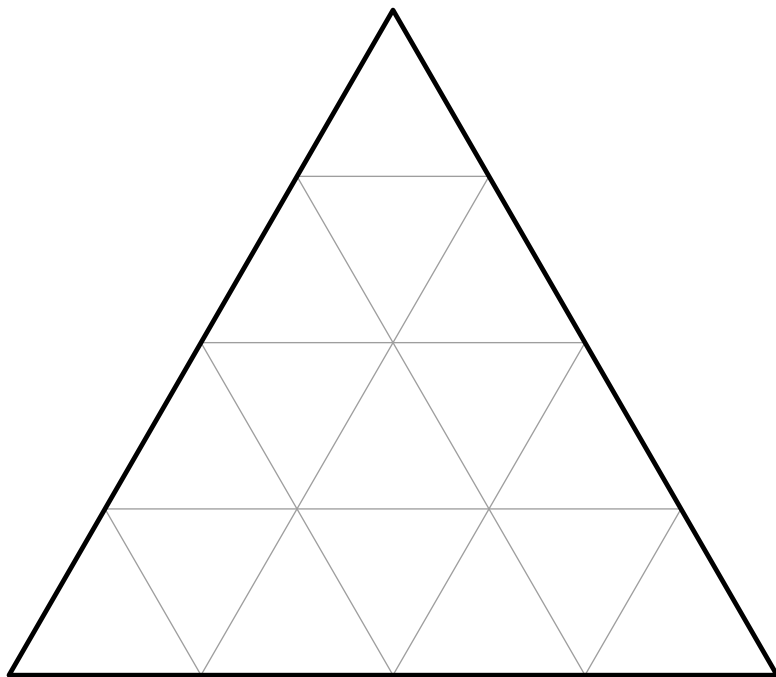
D



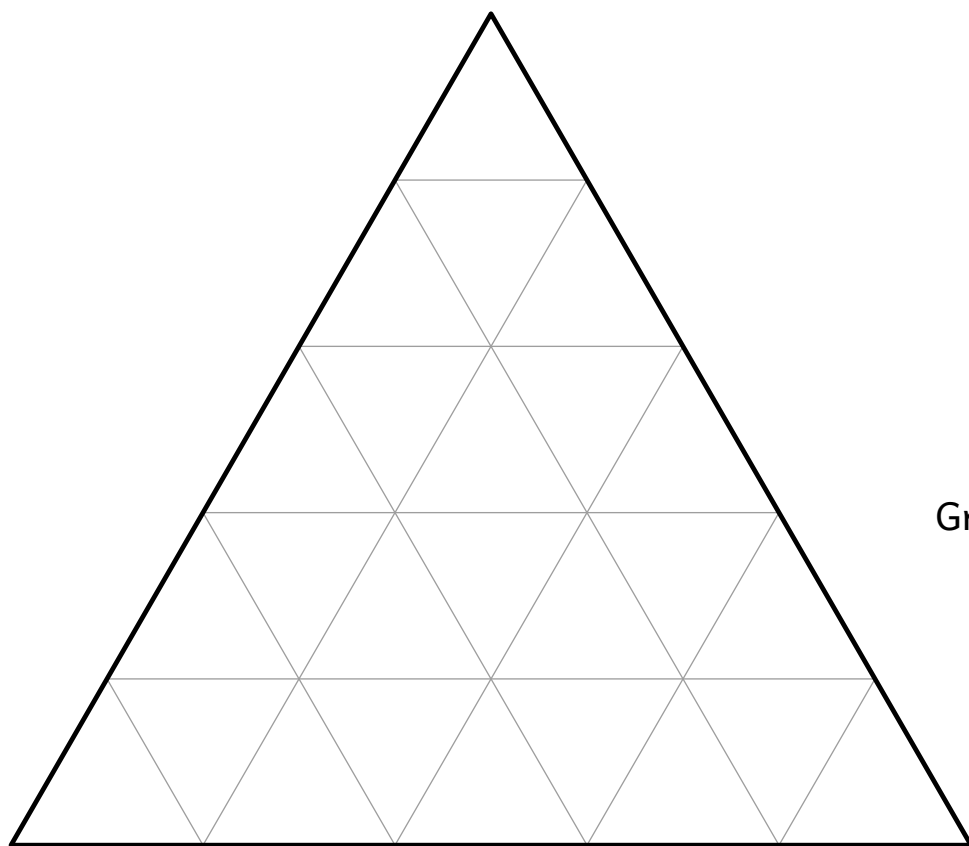
Problem 2: Cover each triangle with blue rhombuses and green triangles. Use as few greens as you can. Shade the greens on the board and write how many.



Greens:



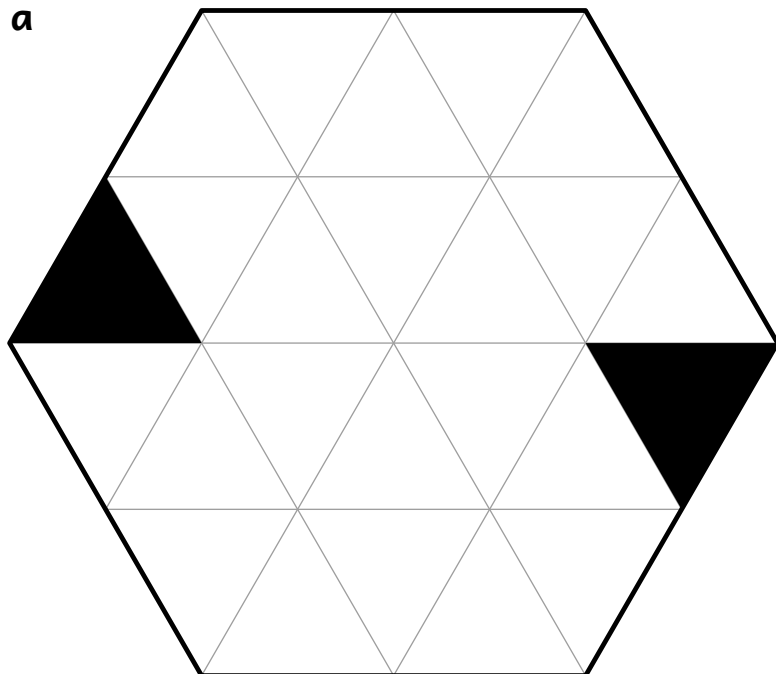
Greens:



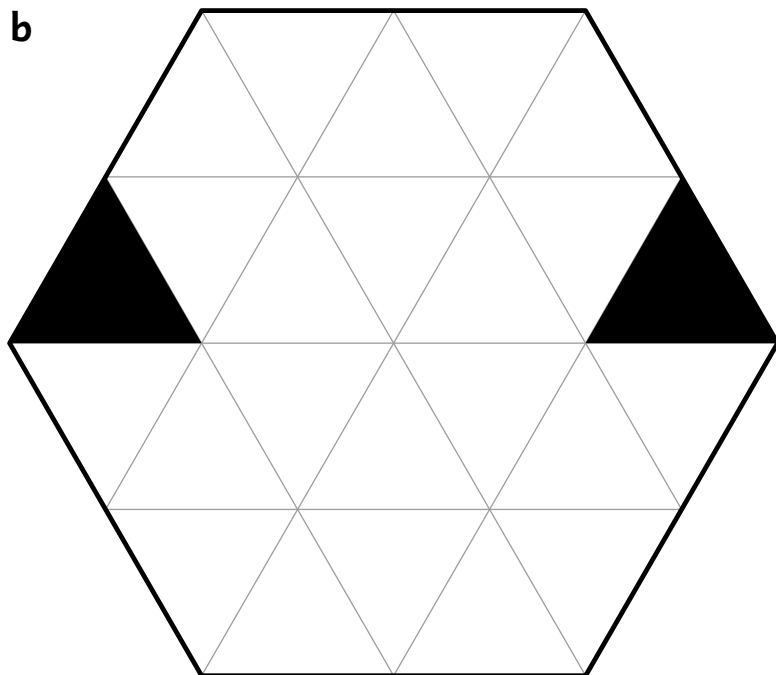
Greens:

Problem 3: Each board is the big hexagon with two small triangles cut out (the black ones). Which boards can be covered with blue rhombuses? Draw the covering on each board that can be covered. For each board that cannot, explain why.

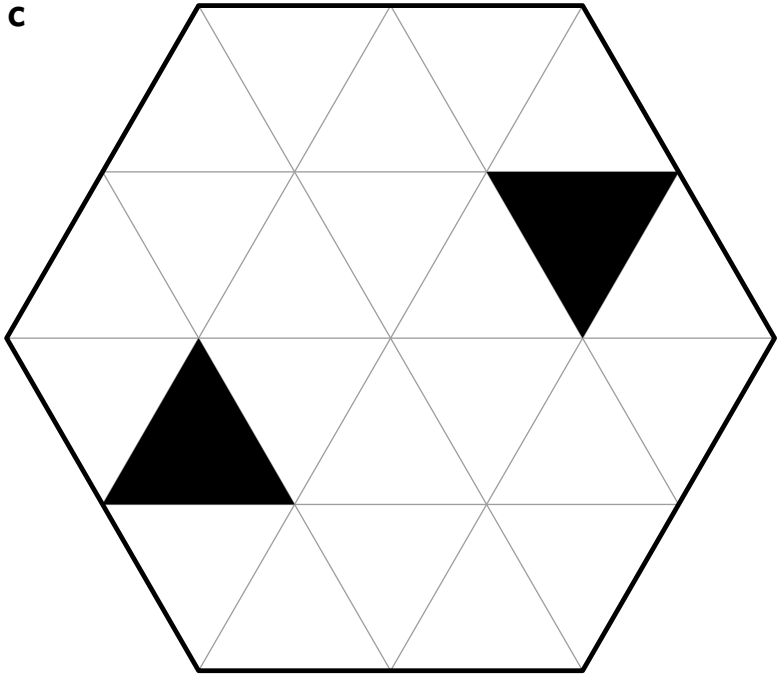
a



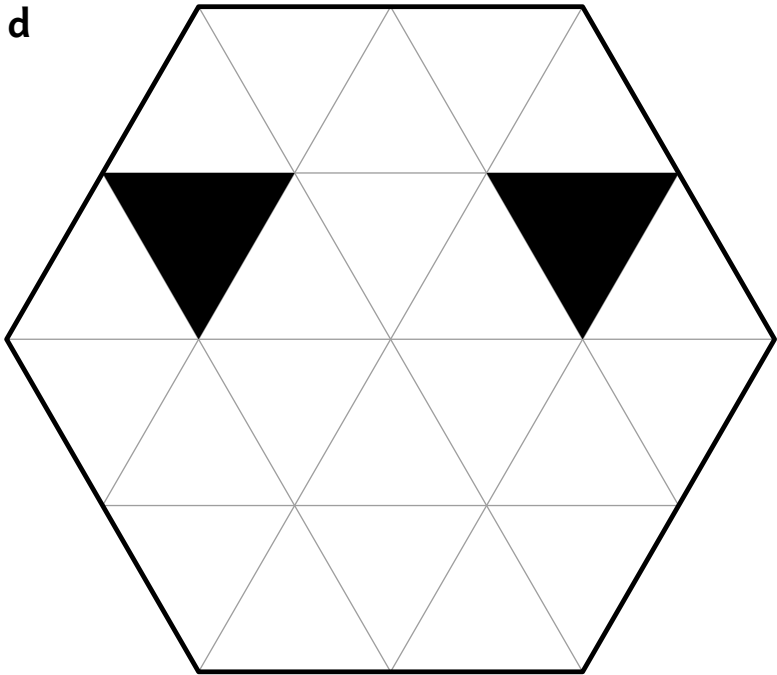
b



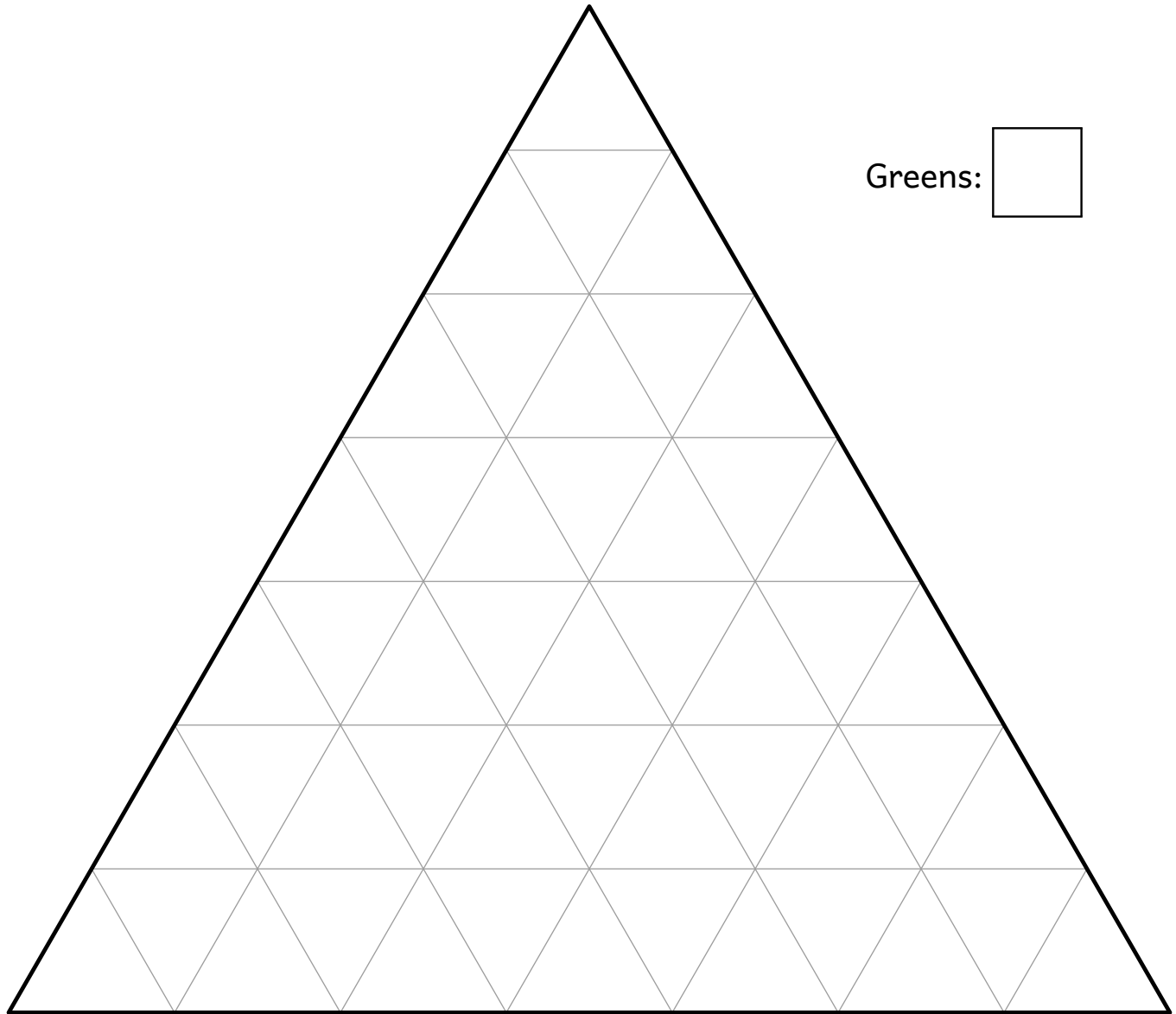
c



d

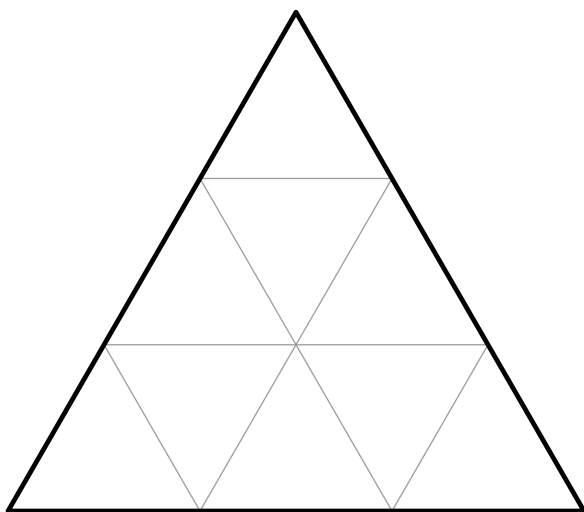
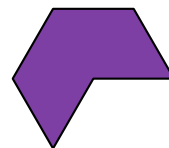


Problem 4: Cover this triangle with blue rhombuses and as few green triangles as you can. Explain why nobody could use fewer greens.

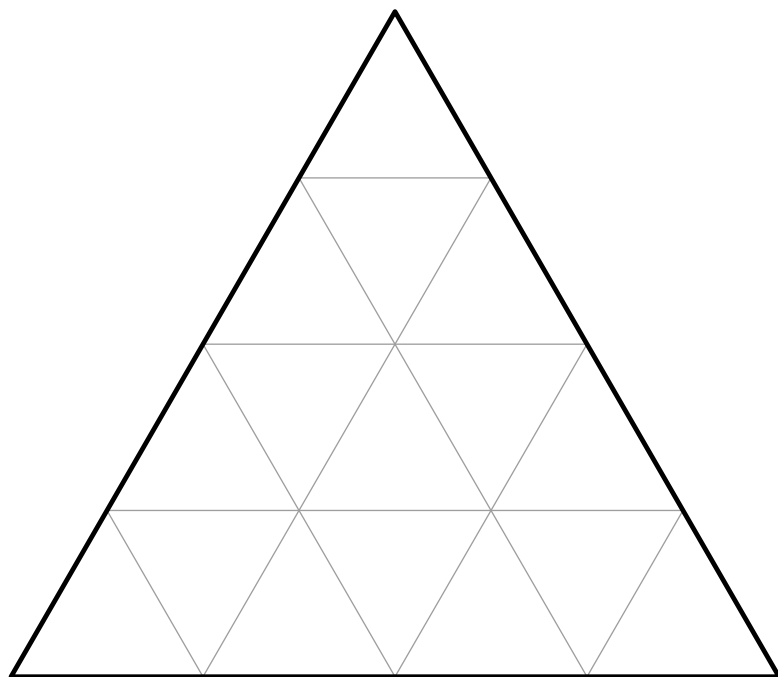


Greens:

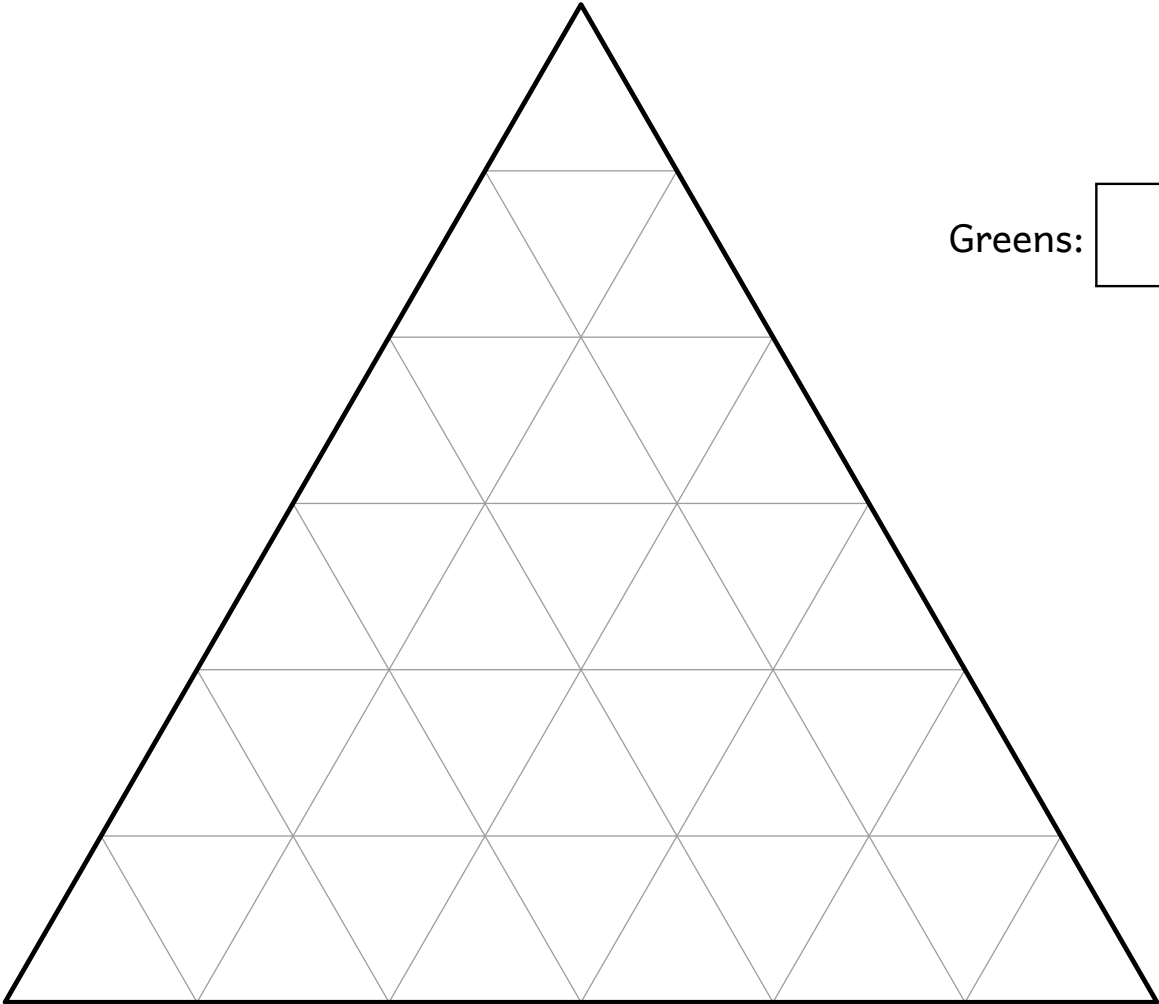
Problem 5: Cover each triangle with purple chevrons and green triangles. Do not use blue rhombuses. Use as few greens as you can, and write how many. Then fill in the table on the next page to compare purple chevrons with blue rhombuses on the same triangles. Could purple chevrons ever need fewer greens than blue rhombuses, on any board at all? Explain.



Greens:



Greens:

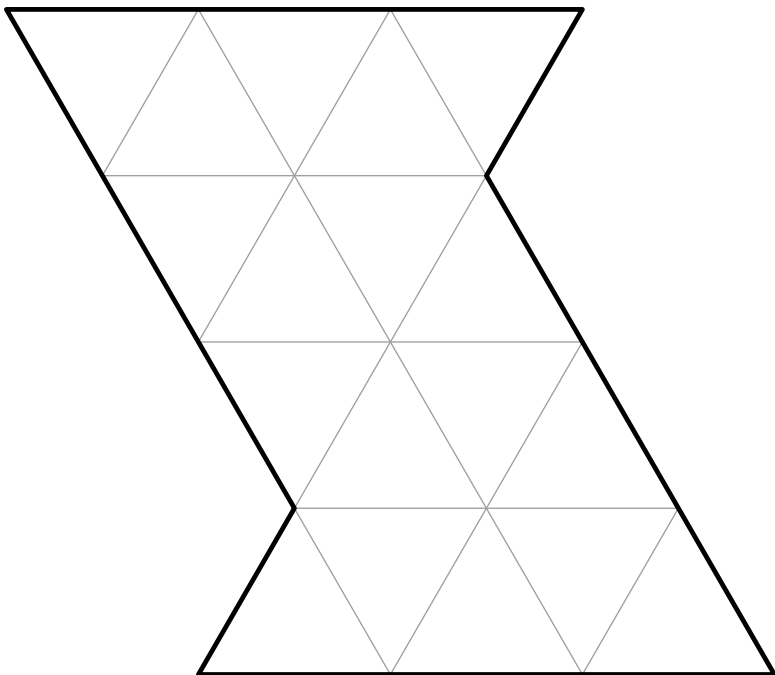


Greens: ☐

Triangle	3 on each side	4 on each side	6 on each side
Greens with blue rhombuses			
Greens with purple chevrons			

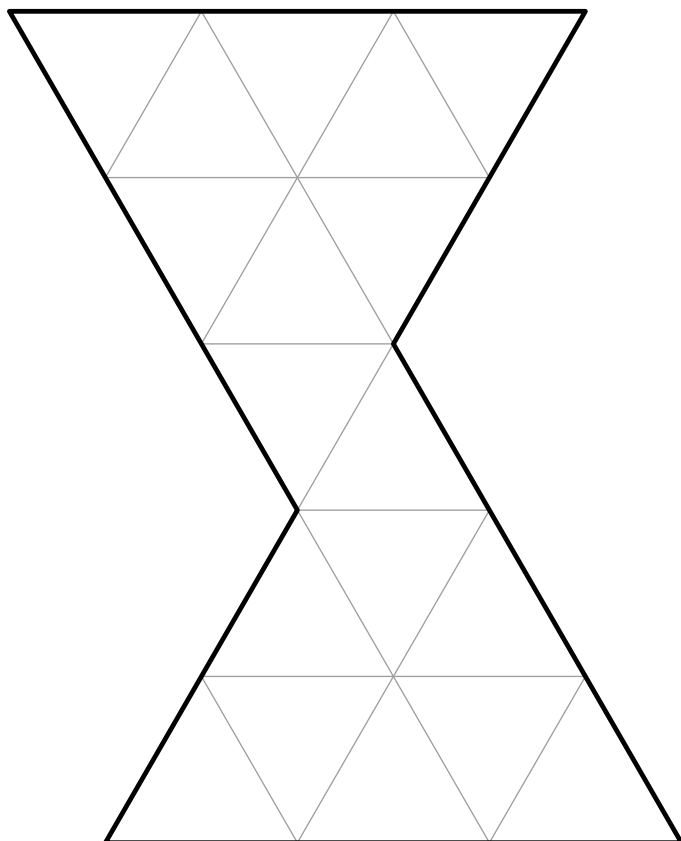
Problem 6: Cover each board with blue rhombuses and as few green triangles as you can. Explain why you cannot use fewer.

a



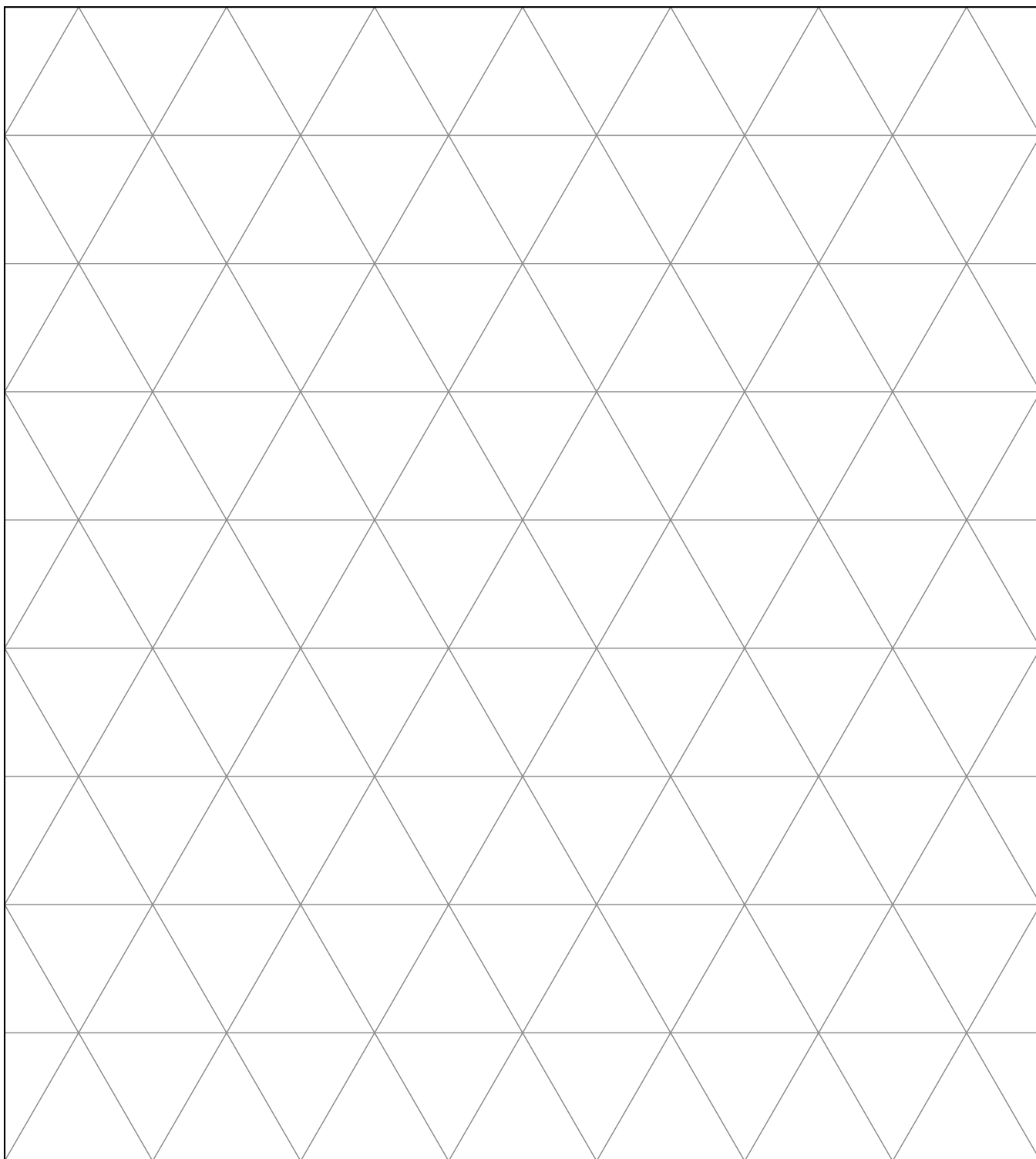
Greens:

b



Greens:

Problem 7: On the grid below, draw a board in one piece, with its small triangles joined along their edges. It must need at least 3 green triangles however you cover it with blue rhombuses and green triangles. Use as few small triangles as you can.



Small triangles in my board: